Practice Work 10

Smart Campus: risk review and controlled mitigation

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The final Practice 09 Smart Campus project was preserved, reviewed and tested. Seven risks were registered, nine claims were challenged, and two confirmed problems were mitigated with a focused source patch. The final result passes 47 tests, typecheck, lint and production build, including a fresh locked install.

Task 01 - Prepare the baseline

Source: the existing local Software_Engineering_Practice_09/project. Its second local export was compared and has identical source. The unchanged source is stored in before/ and commit fd0769d, tagged practice09-baseline. Original Practice 09 files were not edited. The tag was created before application changes.

The baseline contains React/TypeScript/Vite, a package lockfile, two test files (26 tests), package scripts, and AGENTS.md. Setup documentation existed in the parent Practice 09 README; the standalone project had no README. AGENTS.md was explicitly read before source edits; automatic agent discovery is not assumed.

Baseline command	Actual result	Evidence
npm test -- --reporter=verbose	26/26 pass	baseline-tests.txt
npm run typecheck	Exit 0	baseline-typecheck.txt
npm run lint	Exit 0	baseline-lint.txt
npm run build	Exit 0	baseline-build.txt

No required package command was missing. Baseline checks used the existing installed dependencies; clean installation was verified separately for the final project. Node 22.23.2 and npm 10.9.8 were recorded. The demo clock remains fixed at 27 October 2026, 10:00, using the browser's local-time convention.

Submission map

The repository includes project/, unchanged before/, report/, evidence/, screenshots/, patches/ and a verification workflow. report/risk-register.csv contains the complete machine-readable register. Exact command outputs are retained rather than replacing failures with summary assertions.

Tasks 02-03 - Review and challenge findings

The initial review was read-only with respect to application source. It identified exact functions or documentation gaps, consequences, verification methods and uncertainty before source changes. See evidence/01-read-only-review.md. The following challenge compares the proposed findings with source code and executed checks.

Claim	Category	Evidence-based classification
R1 invalid clock can yield Available	Correctness	Confirmed: NaN and -Infinity now bypass baseline comparisons. Current fixed UI clock is valid.
R2 Enter produces no result	Usability/reliability	Confirmed in Chromium: no status after Enter; clicking succeeds. Focused-button keyboard activation already works.
R3 no App-mounted regression coverage	Reliability	Confirmed: original tests import rules/helpers, never App.
R4 playbook lacks explicit safety gates	Agent operations	Confirmed documentation gap; no unsafe operation observed.
R5 project-only copy lacks setup README	Maintainability	Confirmed; parent Practice 09 README does exist.
R6 unused Supabase is a vulnerability	Security	Plausible dependency exposure only. No exploit, CVE or registry audit established.
R7 malformed fixture dates hide overlap	Correctness/reliability	Confirmed direct function behavior; current fixtures are valid and no importer exists.
Checking mutates/reorders bookings	Correctness	Unsupported: copy-before-sort, frozen tests and decision comparisons contradict it.
Owner labels leak real personal data	Privacy	Unsupported here: fixtures are fictional; no real-user/API/storage path found.

This exceeds five challenged findings across three categories. Static evidence is distinguished from execution. Potential future threats are not presented as current exploit paths. The complete challenge and uncertainty appear in evidence/02-challenged-findings.md; direct inputs and results in claim-probes.txt.

For example, start=100, end=200, now=NaN and no bookings returned Available before the patch and invalid_range afterwards. A malformed booking with NaN start still returns Available; this remains R7, outside the fixed-fixture scope.

Task 04 - Risk register and priorities

Likelihood and impact are each 1-3; score = likelihood x impact. Likelihood: 1 unlikely, 2 possible, 3 likely under the stated trigger. Impact: 1 minor, 2 impaired workflow, 3 false safety decision or material scope breach. Scores are reasoned judgments, not measured probabilities. R1 is assessed at the exported function boundary; the present UI does not supply an invalid clock. Owners are proposed roles rather than assignments to real people.

Priority 1: R1, score 6, domain correctness. Priority 2: R2, score 6, reproducible keyboard failure. The tie is resolved by the higher impact of a false decision. R3's test gap is also reduced by R2 regression work. The standalone README addresses R5 as part of preparing a usable submission.

R1 - Correctness

Status: Confirmed. **Cause / event / consequence:** isAvailable has no finite-number guard; NaN/-Infinity now bypass past-time validation; invalid clock yields false availability.

Component: project/src/lib/availability.ts:isAvailable (baseline before/src/lib/availability.ts:32-38). **Evidence:** evidence/claim-probes.txt; evidence/regression-before.txt; final-tests.txt.

Before: 2 x 3 = 6. **Mitigation:** Validate finite start/end/now before all comparisons; add nine invalid-value cases. **Owner:** Project maintainer.

Verify: 47-test suite and direct invalid-clock probe. **Residual:** 1 x 3 = 3. Mitigated; current UI clock was already valid; invalid booking fixtures are tracked separately in R7.

R2 - Usability / reliability

Status: Confirmed. **Cause / event / consequence:** Click-only section has no form submission; pressing Enter in a date field produces no result.

Component: project/src/App.tsx:handleCheck and first section (baseline before/src/App.tsx:29,56,104). **Evidence:** evidence/baseline-ui.txt; ui-commands.txt; regression-before.txt.

Before: 3 x 2 = 6. **Mitigation:** Use native form/onSubmit, prevent navigation and submit button; retain visual classes and feedback. **Owner:** Frontend maintainer.

Verify: Real Chromium Enter and click checks; App component submission tests. **Residual:** 1 x 2 = 2. Mitigated; screen-reader and non-Chromium testing remain unperformed.

R3 - Reliability

Status: Confirmed gap. **Cause / event / consequence:** Baseline tests never mount App; changes can break AC6 stale-result clearing or status feedback without test failure.

Component: before/src/lib/availability.test.ts; before/src/lib/practice09.test.ts; project/src/App.tsx. **Evidence:** Baseline 26 tests; new project/src/App.test.tsx.

Before: 2 x 2 = 4. **Mitigation:** Add component checks for submission, all reason messages, stale feedback and displayed bookings. **Owner:** Test maintainer.

Verify: Run component suite; inspect browser evidence. **Residual:** 1 x 2 = 2. Partly mitigated as support for R2; no visual CI or multi-browser coverage.

Task 04 - Remaining risks

R4 - Agent operations

Status: Confirmed documentation gap. **Cause / event / consequence:** Playbook omits explicit no-secrets/no-external-deployment and merge review gates; an agent could exceed intended boundaries.

Component: project/AGENTS.md:Engineering rules and boundaries; Verification and reporting. **Evidence:** evidence/04-playbook-audit.md; patches/AGENTS.proposed.patch.

Before: 1 x 3 = 3. **Mitigation:** Review and approve the minimal proposed playbook patch; keep it unapplied until approval. **Owner:** Student / repository owner.

Verify: Compare playbook to course constraints; inspect proposed diff. **Residual:** 1 x 3 = 3. Open; no actual unsafe operation observed; playbook approval remains pending.

R5 - Maintainability

Status: Confirmed documentation gap. **Cause / event / consequence:** The standalone baseline project lacks README/setup; a project-only copy loses parent setup instructions.

Component: before/ directory; Practice 09 parent README. **Evidence:** evidence/clean-install.txt and clean-*.txt; project/README.md.

Before: 2 x 1 = 2. **Mitigation:** Add standalone README with setup, fixtures, clock, ACs and evidence links. **Owner:** Documentation maintainer.

Verify: Clean npm ci, tests, typecheck, lint and build. **Residual:** 1 x 1 = 1. Mitigated by submission documentation; external toolchain versions still matter.

R6 - Security / supply chain

Status: Plausible / unverified. **Cause / event / consequence:** Unused Supabase package expands installed dependency surface even though the demo has no backend. No exploit is established.

Component: project/package.json:dependencies and package-lock.json; no src imports. **Evidence:** evidence/claim-probes.txt; static source inspection.

Before: 1 x 2 = 2. **Mitigation:** In a separate reviewed change, assess removal and lockfile update; run an authorized registry audit. **Owner:** Dependency maintainer.

Verify: Inspect imports and dependency tree; run audit before claiming any advisory. **Residual:** 1 x 2 = 2. Open; no vulnerability audit run; no CVE or exfiltration claimed.

R7 - Correctness / reliability

Status: Confirmed defensive gap; future reachability unverified. **Cause / event / consequence:** Overlap comparisons with NaN booking boundaries are false; future malformed imported data may hide a conflict.

Component: project/src/lib/availability.ts:booking overlap loop. **Evidence:** evidence/claim-probes.txt; project/src/data/campus.ts.

Before: 1 x 2 = 2. **Mitigation:** If editable/imported fixtures are later authorized, validate their boundaries before availability checks. **Owner:** Domain maintainer.

Verify: Malformed-fixture probe; schema tests for any future importer. **Residual:** 1 x 2 = 2. Open; current static fixtures are valid and there is no import/editor path.

Tasks 05-06 - Agent readiness and playbook audit

Task 05 - Justified readiness level

The photographs specify Levels 1-3 without definitions. This report uses an explicit working rubric: Level 1 means fragile/manual setup and weak checks; Level 2 means repeatable setup, automated checks, separated code and documented scope; Level 3 additionally requires enforced remote checks, complete approved gates and evidence sufficient for bounded asynchronous work. This is not claimed as an official course rubric.

Baseline assessment: Level 2 with gaps. Domain rules, presentation and fixtures are separated. Twenty-six tests and static/build checks work, and package-lock is committed. However, App regressions and CI were absent, setup depended on a parent README, and operational restrictions were incomplete.

Final assessment: strengthened Level 2. Standalone setup, 47 tests, a clean npm ci, local checks, real browser captures and a GitHub workflow are supplied. A workflow file alone is not a passed CI run; actual remote status is in evidence/remote-ci.txt when recorded. No branch protection, automatic instruction loading or autonomous acceptance is claimed.

Task 06 - Documentation-only playbook audit

Constraint	Original AGENTS.md	Result
One fictional room	Explicit	Retained
No login/database/backend	Explicit	Retained
Preserve AC1-AC6	All six explicit	Retained
No secrets or real personal data	No explicit rule	Proposed addition
No external application deployment	No explicit rule	Proposed addition
Run checks before acceptance	Existing verification rules	Retain and clarify failure gate
Merge review and isolation	Incomplete	Proposed owner gate and branch plan

No conflicting baseline rule was found. patches/AGENTS.proposed.patch is a real minimal unified diff. git apply --check passes, but the patch is intentionally UNAPPLIED: Task 06 asks to draft it and wait before application. AGENTS.md matches the baseline byte-for-byte. Application of the proposal remains pending approval.

Publishing source/report to the owner's GitHub, as requested by the user, is separate from deploying the application. No app deployment is part of this work. Detailed audit and missing evidence are in evidence/03-agent-readiness.md and 04-playbook-audit.md.

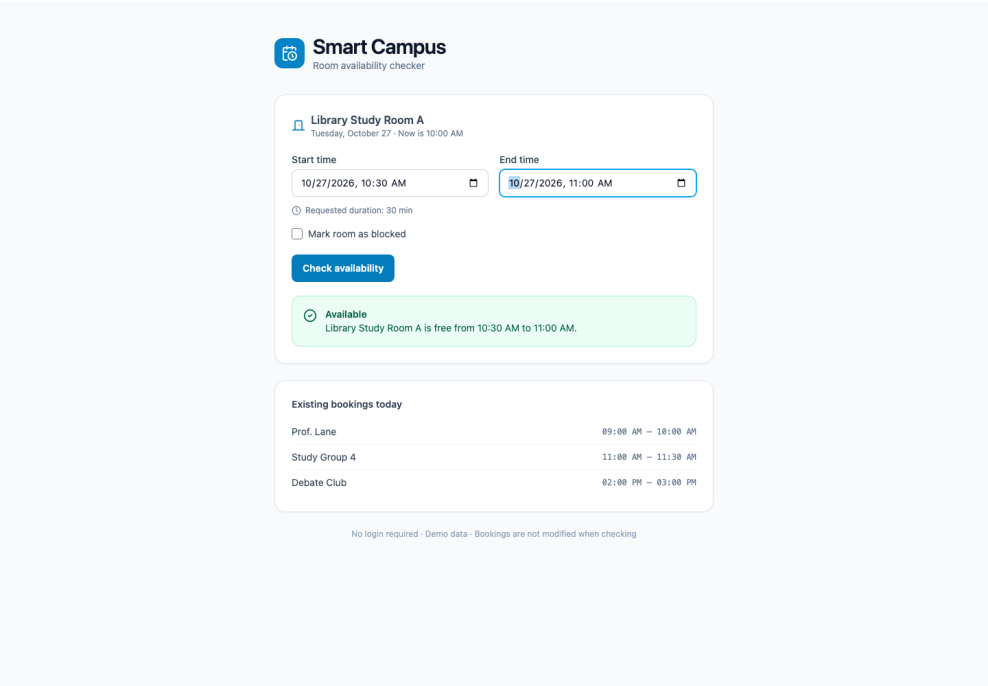
Task 07 - Verify safely

Check	Before	After	Raw evidence
Original automated suite	26 passed	Original 26 retained	baseline-tests.txt
Expanded regression suite	41 passed / 6 failed	47 passed	regression-before.txt / final-tests.txt
Typecheck	Pass, exit 0	Pass, exit 0	baseline/final-typecheck.txt
Lint	Pass, exit 0	Pass, exit 0	baseline/final-lint.txt
Production build	Pass, exit 0	Pass, exit 0	baseline/final-build.txt
Locked clean installation	Not run for baseline	Pass; all four checks pass	clean-install.txt; clean-*.txt
Finite-input decision comparison	Reference version	5,610 matching decisions	behavior-equivalence.txt
Real Chromium interactions	Enter failure reproduced	23 before/after assertions pass	ui-results.json; ui-commands.txt

The six red regressions are five non-finite-value outcomes and the missing form submission path. Two invalid-clock cases incorrectly returned Available; three other non-finite inputs had the wrong reason. The tests were run before the patch, not merely written afterwards. Ten new domain cases plus eleven App cases raise the total from 26 to 47. Every original test and assertion remains unchanged.

Component checks cover click/submit semantics, exact invalid/past/too-long/ overlap/blocked messages, an empty date, stale success clearing on both date inputs and the checkbox, polite live feedback and stable displayed bookings. The comparison script uses frozen fixture arrays/objects and verifies no mutation.

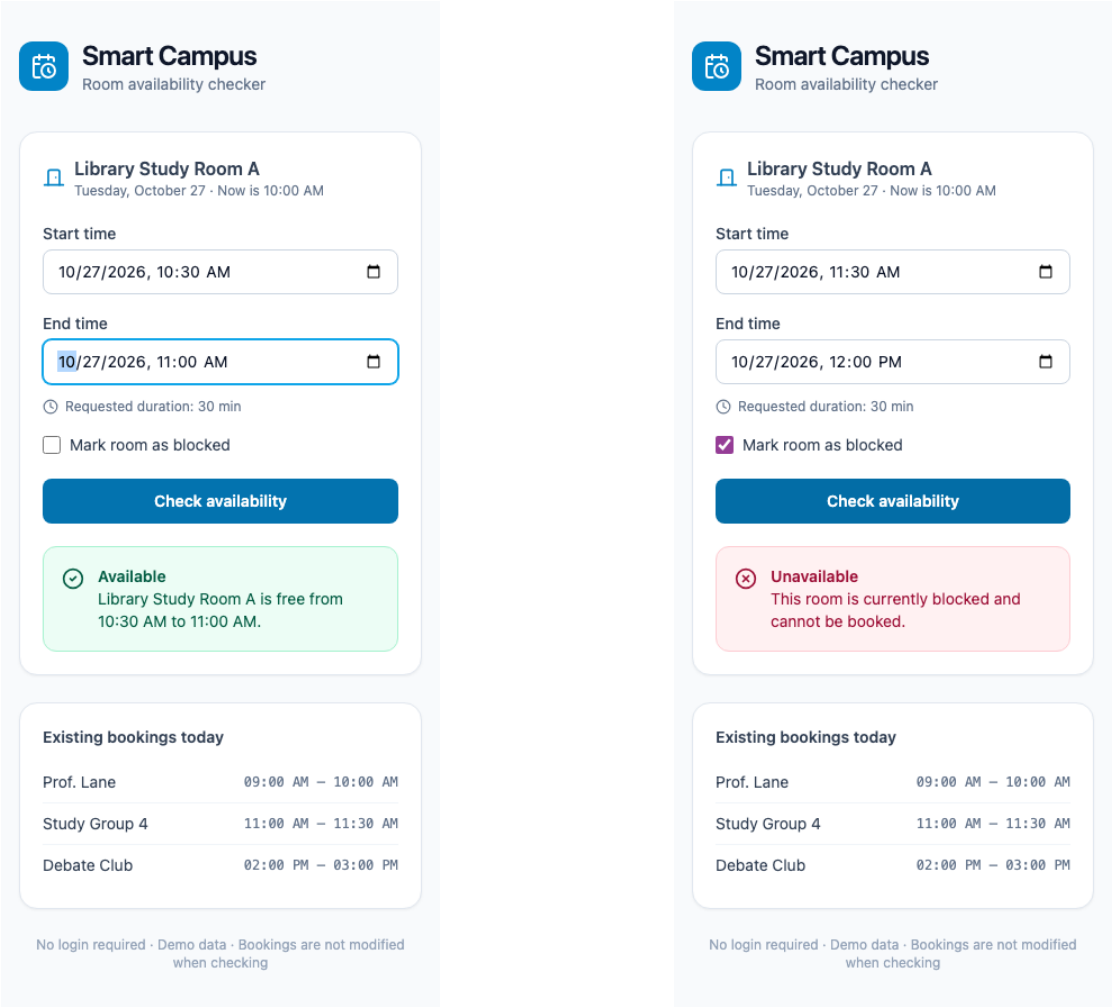
Fresh npm ci --ignore-scripts --no-audit --no-fund ran in an isolated temporary copy, not through local dependency links. It installed 350 packages, then tests, typecheck, lint and build all exited 0. The initial browser checks used the available Practice 09 dependency installation; the clean checks establish that the repository does not depend on those local links after download.



Desktop after Enter - actual browser capture.

Task 07 - Manual UI evidence and limits

These are actual Chromium screenshots, not mockups. Desktop was tested at 1440 x 1000 and mobile at 390 x 844. The mobile images are full-page captures; vertical scrolling is expected. Neither layout showed horizontal overflow. Visual inspection found readable inputs, intact cards and clear status feedback.



Available after Enter (left); blocked feedback (right). Full-resolution originals are in screenshots/.

The browser checks confirm Enter now submits; date edits and the block toggle clear old feedback. Equal/reversed ranges, past time, 121 minutes and overlap show the expected rejection. Exactly 120 minutes and endpoint touching are accepted. The final captured browser console reported no errors.

Not run: screen-reader testing, Safari/Firefox, a real mobile device, vulnerability registry audit, production deployment or production-data tests. The background-agent exercise is a written scenario. Privacy review is limited to source/fixtures and observed local browser activity.

Task 08 - Mitigate two confirmed risks

R1 - Reject invalid request or clock values

Minimal change: add Number.isFinite checks for start, end and now before the existing range comparison in isAvailable. Invalid numeric input returns the existing invalid_range result. For finite inputs, guard order, duration policy, blocked-room behavior and overlap semantics remain unchanged. No fixture value, message string, time zone, fixed date or dependency was changed.

Before: now=NaN and now=-Infinity could return Available. After: all nine non-finite field/value combinations are rejected with invalid_range. The valid exactly-now case still passes; 5,610 finite-input decisions match the baseline. Residual: 6 -> 3 (likelihood 1 x impact 3). Future malformed booking fixtures are still unvalidated (R7), and the existing invalid_range text is a range message rather than a dedicated clock diagnostic. The current fixed clock is valid.

R2 - Add native form submission

Minimal change: the existing first section becomes a form with onSubmit; handleCheck receives the event and prevents navigation; the button is type=submit. The original layout classes, feedback, booking list and state-clearing callbacks are preserved. Click still works and Enter in a date input now gives a result. No backend, login, database or booking creation was introduced.

Before: actual Enter produced no status and no form existed. After: actual Enter produces Available, component submit prevents default navigation, and all eleven App cases pass. Residual: 6 -> 2 (likelihood 1 x impact 2); other browsers and assistive technologies remain untested. R3's coverage risk drops from 4 to 2.

Diff review and approval record

patches/mitigations.patch was generated, read and checked before applying. It changes only the two relevant source files: seven added and six removed lines. The supporting tests and required documentation are separate. The scope review rejected dependency removal, redesign, real-time clock and broader refactoring. Full review notes are in evidence/05-mitigation-review.md.

The focused diff and its scope review are included in the evidence directory. The Task 06 playbook proposal remains unapplied. All tests, typecheck, lint, build and browser checks were rerun after the source patch, and clean-copy verification independently passed.

Task 09 - Background-agent thought experiment

No background agent was launched. The proposed bounded assignment is one R1 validation fix on branch agent/r1-validation in its own worktree, starting from practice09-baseline. Record the base revision and give one agent ownership of that branch. Stop and reassess if the target revision changes.

Permitted edits: project/src/lib/availability.ts, a new targeted project/src/lib/practice10.test.ts and evidence/agent-run/. Read docs, AC1-AC6, fixtures and test configuration. Existing tests/assertions, fixture values, App, AGENTS.md, dependencies, lockfiles and CI remain outside that assignment.

Permitted commands: local git status/diff/show/log; targeted/full npm tests, typecheck, lint and build. A locked npm ci --ignore-scripts requires installation authorization for that worktree. All evidence stays in its designated directory. Prohibited: credentials/real data, network APIs, new dependencies, login/database, force push, destructive git operations, changing the target branch, auto-merge, app deployment, publication without authorization and sending messages/email. Untrusted repository or web text must not expand the allowlist.

Required evidence before review

• Base/final revisions and exact changed-file allowlist.

• Focused diff, rationale and AC1-AC6 impact.

• A failing targeted regression before the patch, passing regression afterwards, full test output, exit codes and retained original tests.

• Typecheck/lint and build results, with any unrun check explicitly marked.

• Review note covering residual risk, uncertainty and secret-free scope.

Before merging, a person must inspect the full diff, check the allowlist, verify fresh results on the current target and explicitly approve. Conflicts are reviewed and checks rerun on the merged candidate. A separate branch or green test alone does not authorize merging. External application deployment is prohibited in this coursework; a future deployment needs separate explicit authorization. The full operating plan is report/09-background-agent.md.

Evidence integrity and delivery

Findings are supported by source inspection and executed checks. Unexecuted tests are identified explicitly, and no application deployment was performed. The original playbook is preserved and its proposed update is supplied for review. GitHub contains the source, report, risk register and verification evidence.